$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi}} e^{\frac{ipx}{\hbar}}.$$

- Putting it all together, we find that

$$\psi(x) = \frac{1}{\sqrt{2\pi}} \int dp e^{\frac{ipx}{\hbar}} \tilde{\psi}(p).$$

Let's take another look at the two equations for going back and forth from position to momentum. Notice how symmetrical they are. The only asymmetry is that one equation contains $e^{\frac{ipx}{\hbar}}$ and the other contains $e^{\frac{-ipx}{\hbar}}$:

$$\begin{aligned} \tilde{\psi}(p) &= \frac{1}{\sqrt{2\pi}} \int dx e^{\frac{-ipx}{\hbar}} \psi(x) \\ \psi(x) &= \frac{1}{\sqrt{2\pi}} \int dp e^{\frac{ipx}{\hbar}} \tilde{\psi}(p). \end{aligned} \tag{8.25}$$

The relation between the position and momentum representations summarized by Eqs. 8.25 is that they are reciprocal *Fourier transforms* of one another. In fact, these are the central equations in the field of Fourier analysis. I want you to notice how easy it was to derive those equations using Dirac's elegant notation.

8.4 COMMUTATORS AND POISSON BRACKETS

Earlier, in Lecture 4, we formulated two important principles about commutators. The first had to do with the connection between classical mechanics and quantum mechanics; the second had to do with uncertainty. I